

Adversarial Transferability of Deep Learning-Based LPI Radar Detectors: Comparative Study of CNN and Transformer Architectures

Moohyun Lee*, Sungjae Park, Sunghwan Cho, Yee Wei Law, Sang-Heon Lee,

Yongchul Kim[°]

ABSTRACT

Recent advances in detecting and classifying low-probability-of-intercept (LPI) radar signals have increasingly relied on deep learning techniques operating on time–frequency images (TFIs). CNN-based detectors, in particular, have achieved high accuracy; however, their robustness to adversarial perturbations remains insufficiently studied. This paper extends the analysis to systematically evaluate the impact of model architecture on adversarial effectiveness. We conduct a comparative study across CNN and Transformer architectures to quantify how structural differences in the detector influence attack outcomes. Furthermore, we explore ensemble-based adversarial generation methods that combine multiple surrogate models to produce transferable perturbations across heterogeneous detector types. Experimental results show that adversarial examples generated from a single architecture are often architecture-specific, while those crafted through multi-model ensembles achieve higher attack success rates (ASR) and broader generalization. These findings highlight the importance of architecture-aware adversarial evaluation and demonstrate that ensemble-based attack strategies pose a more significant threat to deep-learning-based radar detection systems.

Keywords: LPI radar, Adversarial attack, Transferability, Ensemble methods, Time–Frequency Image

I. Introduction

Low-Probability-of-Intercept (LPI) radar signals are designed to minimize the likelihood of detection by enemy receivers while maintaining reliable surveillance, tracking, and communication capabilities. In modern warfare, where dominance in the electromagnetic spectrum directly translates to tactical and strategic superiority, LPI radars are a cornerstone technology for achieving stealth, survivability, and information dominance. They enable military platforms—such as aircraft, ships, and missiles—to conduct covert

reconnaissance, precision targeting, and situational awareness without compromising their position through radar emissions. As adversaries continue to develop increasingly sensitive electronic support measures (ESM) and interception systems, the sophistication of LPI waveforms has evolved dramatically. Modern LPI systems employ frequency agility, phase coding, and spread-spectrum techniques to conceal their energy footprint, creating waveforms that are intentionally difficult to detect or classify even with advanced signal processing tools. Consequently, the ability to identify and classify LPI signals under low

* First Author : Department of Electronic Engineering, Korea Military Academy, mh6473@kma.ac.kr, Professor, Member

[°] Corresponding Author : Department of Electronic Engineering, Korea Military Academy, kyc6454@gmail.com, Professor, Member

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signal-to-noise ratio (SNR) conditions has become an essential capability in electronic warfare (EW) and spectrum sensing operations [1].

To address this challenge, deep learning-based LPI radar detection methods have recently attracted substantial attention. In particular, models operating on time-frequency images (TFIs) have demonstrated strong performance in representing and distinguishing LPI modulations by capturing their spectro-temporal structures. Detectors based on convolutional neural networks (CNNs) achieved high classification accuracy and generalization when trained on TFI datasets, often outperforming conventional feature-engineered approaches [2-5]. However, despite their high accuracy under clean conditions, deep learning-based signal classifiers are known to be vulnerable to adversarial perturbations, in which small, intentionally crafted input modifications can induce incorrect predictions [6-9].

Adversarial vulnerability has also been investigated in the radio-frequency and modulation-classification domains. Previous studies have demonstrated that adversarial perturbations can substantially degrade deep learning-based radio signal classifiers under both white-box and black-box settings [9], and subsequent studies have investigated transferable attacks designed to improve attack effectiveness against unknown modulation classifiers [10-11]. Other studies have considered more practical constraints, including wireless-channel effects, shift invariance, and the preservation of signal characteristics during adversarial generation [11-13]. In the LPI radar domain, previous studies have evaluated the vulnerability of TFI-based LPI detectors to gradient-based attacks and demonstrated that adversarial perturbations generated using surrogate models can transfer to independently trained target detectors [13-14]. Collectively, these studies establish that adversarial vulnerability and transferability are relevant

concerns for deep learning-based signal recognition systems.

Nevertheless, the influence of detector architecture on adversarial transferability in LPI radar recognition remains insufficiently characterized. Existing studies have primarily focused on demonstrating adversarial vulnerability, developing individual attack-generation techniques, or evaluating transferability within a limited set of detector structures. In particular, systematic comparisons of transfer behavior between CNN- and Transformer-based LPI detectors remain limited. Furthermore, it remains unclear whether combining structurally heterogeneous surrogate models can generate perturbations that generalize more effectively to independently configured target detectors. This question is particularly relevant to black-box scenarios, where an attacker may construct surrogate models based on available knowledge but does not have access to the exact architecture, trained parameters, or gradients of the deployed detector.

Accordingly, this study focuses on the architecture-dependent transferability of adversarial examples rather than merely demonstrating the adversarial vulnerability of an individual LPI detector. We investigate four representative architecture families: two CNN-based models, ResNet50 [15] and MobileNetV2 [16], and two Transformer-based models, ViT-B-16 [17] and Swin-S [18]. We first quantify single-model transferability within and across these architecture families and subsequently investigate whether heterogeneous surrogate ensembles improve transferability toward independently configured detector models.

This study therefore addresses two fundamental research questions: (i) To what extent do architectural differences between CNN- and Transformer-based LPI detectors influence adversarial transferability? (ii) Can adversarial generation using heterogeneous surrogate ensembles improve transferability to

unseen detector models under black-box conditions?

To answer these questions, this paper makes the following key contributions:

- Cross-architecture transferability analysis: We systematically evaluate adversarial transferability across CNN-based (ResNet50 [15], MobileNetV2 [16]) and Transformer-based (ViT-B-16 [17], Swin-S [18]) LPI radar detector architectures, enabling both within-family and cross-family comparisons of adversarial vulnerability.
- Heterogeneous ensemble-based attack analysis: We investigate adversarial generation using multiple structurally different surrogate architectures and quantify how surrogate composition and architectural diversity affect transferability across detector families.
- Black-box evaluation using independently configured target detectors: For ensemble-based evaluation, the target detectors are constructed and trained independently from the surrogate models, with architectural configurations varied within each detector family. The reported results therefore evaluate whether perturbations generated from fixed surrogate models generalize to unseen detector instances whose exact configurations, parameters, and gradients are unavailable to the attacker.

Through these experiments, we show that adversarial transferability is strongly architecture-dependent and that perturbations generated from individual surrogate models often exhibit limited cross-architecture generalization. In contrast, heterogeneous surrogate ensembles can improve attack transferability across independently configured detector models. These findings highlight the importance of architecture-aware robustness evaluation when assessing adversarial threats to deep learning-based LPI radar recognition

systems.

The remainder of the paper is organized as follows. Section II outlines the LPI signal set, TFI preprocessing, and detector architectures. Section III presents the adversarial threat model, attack algorithms, and ensemble-based generation strategy. Section IV presents the experimental results and analyzes single-model and ensemble-based transferability across detector architectures. Section V summarizes the main findings, discusses the limitations of TFI-domain adversarial analysis, and outlines directions for future research.

II. Deep Learning-Based LPI Detection Model

2.1 Signal Set

LPI radars are designed to minimize their likelihood of detection by adversarial receivers, often using complex waveform structures and emission strategies. Modern systems adopt a variety of modulation techniques, carrier frequencies, and amplitude profiles—while maintaining low power levels—to further reduce intercept probability. These waveforms are typically characterized by low peak power, wide spectral occupancy, narrow directional beams, and subtle intra-pulse modulation, which collectively undermine the effectiveness of conventional spectrum-based detectors such as those relying on the Fast Fourier Transform (FFT).

In this work, the received LPI radar signal is represented in complex baseband form contaminated by additive noise [1]:

$$x(t) = Ae^{j(2\pi f_c t + \phi(t))} + n(t) \quad (1)$$

where A is the signal amplitude, f_c is the carrier frequency, $\phi(t)$ denotes the intra-pulse phase modulation, and $n(t)$ is complex additive white Gaussian noise. This formulation reflects the typical signal observed at the intercept receiver and forms the basis for

time-frequency analysis and classification. While this parametric model describes signal structure at a high level, practical classification performance is largely dictated by signal features tied to the ambiguity function, autocorrelation, and other time-frequency characteristics governed by the modulation scheme.

LPI radar signals are broadly categorized based on the nature of their intra-pulse modulation. Frequency-modulated (FM) waveforms dynamically adjust their instantaneous frequency within the pulse duration, while phase-modulated (PM) waveforms alter the phase content of the signal without varying the carrier frequency. FM waveforms include techniques such as linear frequency modulation (LFM) and Costas codes, which introduce time-varying frequency sweeps or hops. In contrast, PM waveforms are often constructed using binary or polyphase codes, including BPSK, Frank, and the P1-P4 and T1-T4 code families. These schemes produce distinctive patterns in the time-frequency domain that can be exploited by machine learning models for classification. Following established benchmarks in LPI waveform recognition, this study focuses on twelve representative modulation schemes: two frequency-based (LFM, Costas) and ten phase-based (BPSK, Frank, P1, P2, P3, P4, T1, T2, T3, T4). These waveforms are widely used in prior research for evaluating detection and classification performance. For an in-depth technical overview of these modulation formats and their operational significance, the reader is referred to [1].

The dataset was synthetically generated at a sampling frequency of 3 kHz. The number of IQ samples in each signal was varied from 800 to 1,600. In addition to the SNR and carrier frequency, modulation-specific parameters were randomly varied during signal generation, including code lengths and sequences, bandwidths, frequency-hopping patterns, numbers of subcodes or segments, and other

waveform-specific parameters. The carrier frequency was randomly selected within the range $f_s/6$ to $f_s/5$. This parameter randomization provides substantial intra-class variability while preserving the characteristic structure of each modulation scheme.

The training, validation, and test datasets were independently generated using different random seeds rather than being obtained by partitioning a single generated dataset. For each of the twelve modulation classes, 1,000,000 training samples, 200,000 validation samples, and 100,000 test samples were generated, resulting in 12,000,000, 2,400,000, and 1,200,000 samples, respectively. The training dataset covered SNRs from -16 to 10 dB in 2-dB increments, whereas the validation and test datasets covered -10 to 10 dB in 2-dB increments. For each generated signal, the SNR was randomly selected from the corresponding discrete SNR set.

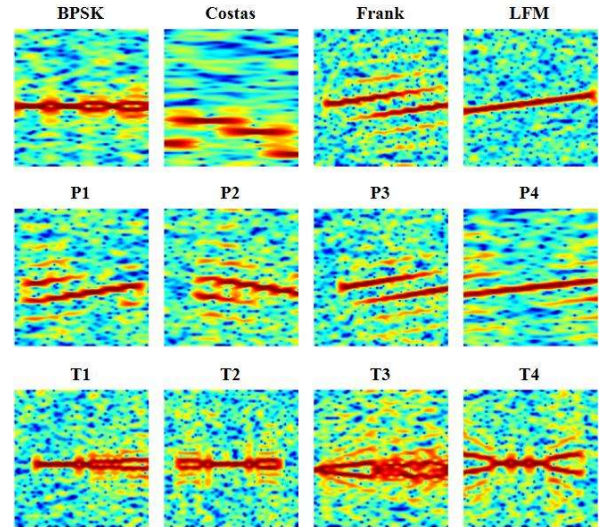


Fig. 1 Sample STFT Time-Frequency Images for Twelve LPI Radar Modulations.

2.2 Time-Frequency Image Generation

To facilitate modulation classification, intercepted LPI radar signals must be converted into a representation that highlights their unique temporal and spectral features. In this study, each complex baseband signal is transformed into a two-dimensional TFI using the Short-Time Fourier Transform (STFT),

which provides a practical balance between resolution and computational efficiency. The continuous STFT of a signal $x(t)$ is defined as:

$$STFT(w, \tau) = \int_{-\infty}^{\infty} e^{-j\omega t} w(t - \tau) x(t) dt \quad (2)$$

where $w(t - \tau)$ denotes the window function that specifies the range of frequency transformation. Accordingly, $STFT(w, \tau)$ represents the Fourier transform of a localized segment of the signal $x(t)$, extracted using the window function $w(t)$ centered at τ . By discretizing the continuous STFT, the following expression is obtained:

$$STFT(l, w) = \sum_{n=-\infty}^{\infty} x[n] w[n - l] e^{-j\omega n}$$

where $w[n]$ denotes the discrete window function and l represents the temporal frame index.

For TFI generation, the original complex IQ signals sampled at $f_s = 3k\text{Hz}$ were directly transformed using the STFT. A Hann window with a length of 128 samples was used with 127-sample overlap, corresponding to a hop length of one sample. The FFT size was set to 128. The magnitude of the complex STFT coefficients, $|X|$, was used to construct the TFI without one-sided spectral truncation. Each TFI was independently min-max normalized to the range $[0, 1]$, converted to an 8-bit grayscale representation, and resized to 224×224 pixels. The resulting single-channel TFIs were used as inputs to the deep learning-based detectors.

This preprocessing technique offers a standardized and visually interpretable representation across all LPI modulations, revealing informative intra-pulse structures such as linear chirps, discrete frequency jumps, and phase-coded textures. These patterns differ significantly between modulation classes, providing rich visual cues for deep learning models to exploit. The TFIs generated from the twelve modulation schemes used in this work are illustrated in Fig. 1. Each image encodes the time-frequency energy distribution of the corresponding signal and displays a

class-specific structure that enables accurate classification by CNNs, Transformers, and other vision-based architectures.

2.3 Detector Architectures

To systematically analyze architecture-dependent adversarial vulnerability and ensemble-based transferability, we select four representative deep learning-based LPI detectors: ResNet50, MobileNetV2, ViT-B-16, and Swin-S [15-18]. These models represent two major families of vision architectures, CNNs and Transformers, and were chosen to cover two axes of variation: model complexity and computational efficiency.

Specifically, ResNet50 and MobileNetV2 are CNN-based detectors. ResNet50 is a high-capacity (heavyweight) model that leverages deep residual connections, while MobileNetV2 is a lightweight architecture that employs depthwise separable convolutions to improve efficiency. By contrast, ViT-B-16 and Swin-S are Transformer-based detectors: ViT-B-16 models global dependencies through patch-level self-attention, whereas Swin-S adopts a hierarchical, windowed attention mechanism that improves scalability and computational efficiency.

All detector models were trained from scratch using the same general training protocol. Cross-entropy loss was optimized using AdamW with an initial learning rate of 1×10^{-3} , a weight decay of 1×10^{-4} , and a batch size of 256. Training was performed for up to 100 epochs. A three-epoch learning-rate warm-up was followed by cosine learning-rate decay. Applying a consistent training protocol across the detector architectures reduces the influence of differences in training conditions when comparing architecture-dependent adversarial transferability.

For adversarial evaluation, the models used for adversarial-example generation and the target detectors used for transfer evaluation are treated separately. In the ensemble

experiments, four fixed surrogate models corresponding to ResNet50, MobileNetV2, ViT-B-16, and Swin-S are used on the attack side. The attacker can access these surrogate models to compute the gradients required for adversarial-example generation but does not have access to the parameters or gradients of the target detectors.

To evaluate transferability beyond the exact surrogate models, five target variants were independently constructed and trained for each detector family. The target variants retain the general characteristics of their corresponding architecture family while varying internal architectural configurations within predefined ranges. For example, the ResNet-family variants differ in the number of residual blocks, convolutional channel configurations, and activation functions. Consequently, the target detectors are not identical copies of the corresponding surrogate models. The ensemble-transfer results for each detector family are reported as the average over the five independently configured target models.

Importantly, even when an ensemble contains a surrogate belonging to the same architecture family as the target detector, the actual target detector itself is not included in the attack-generation process. Its exact architectural configuration, trained parameters, and gradients remain unavailable to the attacker. We therefore distinguish between inclusion of a surrogate from the same architecture family and inclusion of the actual target model when interpreting the ensemble-transfer results.

Our experiments follow two complementary evaluation paradigms. (i) Single-model evaluation: adversarial examples generated from a single attack model are evaluated in both self-model and off-diagonal settings to establish a baseline for architecture-dependent transferability. (ii) Ensemble-based black-box evaluation: adversarial examples are generated using combinations of two, three, or four fixed surrogate models and subsequently evaluated

against the independently configured target detectors described above. The surrogate composition is systematically varied among CNN-only, Transformer-only, and mixed CNN-Transformer combinations to examine the effects of ensemble size and architectural diversity.

Building upon the above, our experiments specifically investigate: (i) Which single surrogate models exhibit relatively higher transferability across heterogeneous architectures; (ii) How ensemble-based generation improves average ASR; (iii) Which ensemble configuration achieves the highest ASR and transferability across different model architectures.

This approach goes beyond identifying the vulnerability of individual detector architectures by examining whether adversarial perturbations generated from fixed surrogate models generalize to independently configured target detectors. Section III formally defines the threat model and describes the adversarial attack algorithms, ensemble-generation strategy, and evaluation metrics.

III. Adversarial Attack Scenarios

Deep learning-based LPI radar detectors, although highly effective under clean conditions, are known to be susceptible to small, well-crafted perturbations known as adversarial examples. These perturbations are intentionally designed to alter model predictions while remaining bounded in the input TFI domain. In this section, we describe the threat model, attack algorithms, and ensemble strategies adopted in our experiments.

We consider a transfer-based black-box attack scenario in which the adversary has no access to the trained parameters or gradients of the target detector and does not know its exact architectural configuration. Instead, the attacker constructs one or more independently trained surrogate models using data drawn

from the same underlying signal distribution and generates adversarial TFIs using only the gradient information available from these surrogate models. The generated adversarial examples are subsequently evaluated on the target detectors without using target-model gradients or feedback to update the attack.

Importantly, black-box transferability in this study does not require the surrogate and target detectors to belong to different broad architecture families. As described in Section II, a target detector may belong to the same architecture family as one of the surrogate models while differing in its detailed architectural configuration and independently learned parameters. Therefore, we distinguish between the architecture family (e.g., ResNet or ViT) and the exact instantiated detector model. In the ensemble-based evaluation, the exact target detector is unseen during adversarial generation, even when a surrogate from the same architecture family is included in the ensemble.

Under this threat model, we investigate whether adversarial perturbations generated from fixed surrogate models generalize to independently configured and trained target detectors, and how such transferability changes according to the architectural relationship between the surrogate and target models.

To evaluate the vulnerability of deep learning-based LPI radar detectors, we implement and compare four representative gradient-based adversarial attack methods: FGSM [6], PGD [7], MI-DI-FGSM (Momentum Iterative Diverse Input FGSM), and MI-DI-PGD (an experimental variant applying both momentum and input diversity to the PGD framework) [20–21]. All four approaches generate adversarial examples by exploiting the gradient of the loss function with respect to the model input, thereby perturbing the original TFIs in directions that maximize classification error under a bounded distortion constraint.

The FGSM serves as a baseline, applying a single-step perturbation proportional to the

sign of the gradient. The PGD method extends gradient-based adversarial generation into an iterative framework, performing multiple gradient steps while projecting the perturbed input back into the allowed perturbation space. To improve stability and transferability, the MI-DI framework is integrated into both FGSM and PGD variants. The momentum term accumulates gradient information from previous iterations to stabilize the update direction, while the input diversity module introduces random spatial transformations—such as random resizing, cropping, or padding—before each gradient computation. These mechanisms help to avoid overfitting to a specific model structure and enhance the black-box transferability of the generated adversarial examples [20–21].

The norm constraint was defined according to the underlying attack formulation rather than uniformly across all four methods. FGSM employed an L_∞ -bounded perturbation, where ϵ specifies the maximum absolute perturbation applied to each normalized TFI pixel. In contrast, PGD employed an L_2 -bounded iterative perturbation, where the adversarial perturbation was projected onto an L_2 ball of radius ϵ after each update. The corresponding MI-DI variants followed the norm constraint of their respective base attacks; thus, MI-DI-FGSM used an L_∞ constraint, whereas MI-DI-PGD used an L_2 constraint.

For the iterative attacks, the number of iterations was fixed at $T = 20$. The perturbation budget was varied as $\epsilon \in \{2/255, 4/255, \dots, 16/255\}$, and the step size was set to $\alpha = \epsilon / T = \epsilon / 20$. For the momentum iterative variants, the decay factor was set to $\mu = 1.0$, while the input diversity transformation was applied with probability $p = 0.5$. For PGD-based attacks, the perturbation was projected onto the corresponding L_2 ball after each iterative update.

To further improve the transferability and transformation robustness of the generated adversarial examples, we apply the Expectation

Over Transformation (EOT) technique [8]. EOT estimates the attack gradient by averaging gradients obtained from multiple transformed realizations of the input. In our implementation, $N_{EOT} = 10$ transformed realizations were used for each gradient estimation. This procedure reduces the dependence of the generated perturbation on a particular input realization and promotes transferability under minor input transformations.

It should be noted that the adversarial perturbations considered in this study are generated directly in the TFI domain. Therefore, a bounded perturbation in TFI pixel space does not necessarily correspond to a uniquely realizable perturbation of the original IQ waveform. Establishing an explicit mapping between the TFI-domain perturbation magnitude and physically realizable waveform-domain interference is beyond the scope of this study. Instead, the present work focuses on the input-space vulnerability and cross-model transferability of TFI-based LPI detectors. The perturbation range was empirically selected to examine adversarial behavior without introducing excessively large distortions that trivially collapse the classifier in the self-model setting. The physical realizability and signal-domain characteristics of TFI-domain adversarial perturbations remain an important direction for future investigation.

After comparing the four attack algorithms under the same experimental conditions, the method achieving the highest overall ASR was selected as the common attack method for the subsequent transferability experiments. Using a common attack method allows the effects of surrogate architecture and ensemble composition to be examined independently of differences among attack algorithms. We first evaluate single-model attacks using ResNet50, MobileNetV2, ViT-B-16, and Swin-S to characterize self-model attack effectiveness and off-diagonal transferability across detector architectures. We then extend the analysis to ensemble-based adversarial generation using

combinations of the four fixed surrogate models. The ensemble attacks are evaluated against the independently configured target detectors described in Section II, allowing transferability to be assessed without using the exact target detector during adversarial generation. By varying both the number and architectural composition of surrogate models, we investigate how surrogate diversity influences transferability across detector families.

Various ensemble aggregation strategies for improving the transferability of adversarial examples have been widely investigated in the literature [19, 22]. Among these methods, we adopt the logit-mean ensemble strategy in which the pre-softmax logits of multiple surrogate models are averaged before computing the adversarial loss. Let $z_x(x) \in R^C$ denote the logits of the k -th surrogate model for input x . For an ensemble consisting of K surrogate models, the ensemble logits are defined as

$$\overline{z_{ens}}(x) = \frac{1}{K} \sum_{k=1}^K z_k(x),$$

The loss for adversarial optimization is then computed as $L_{ens}(x, y) = l_{CE}(\overline{z}(x), y)$, where l_{CE} is the cross-entropy loss. The input gradient used for generating perturbations is:

$$g(x) = \nabla_x L_{ens}(x, y)$$

Here, L_{ens} denotes the ensemble loss, defined as the cross-entropy computed on the averaged logits $\overline{z_{ens}}(x)$ from the K surrogate models. Thus, the ensemble attack does not independently generate and subsequently average adversarial perturbations from individual surrogate models. Instead, a single adversarial gradient is obtained from the loss computed on the averaged surrogate logits. This formulation encourages the resulting perturbation to reflect decision characteristics shared across multiple surrogate architectures and thereby improve cross-model transferability.

IV. Experiments and Analysis

This section presents empirical evaluations of

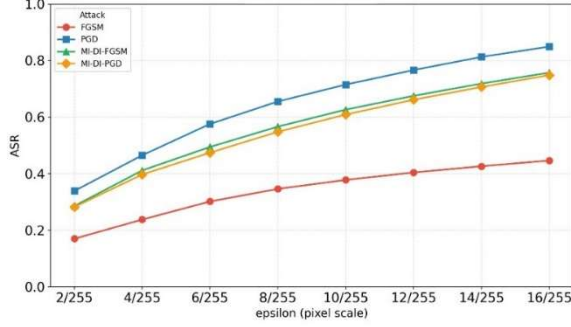


Fig. 2. Average ASR of four adversarial methods under self-model attacks across varying ϵ levels.

adversarial attack performance against deep learning-based LPI radar detectors. The experiments aim to assess the effectiveness and cross-architecture transferability of adversarial examples across different detector models, and to examine the benefits of ensemble-based attack generation.

We evaluate four representative detector architectures, namely ResNet50, MobileNetV2, ViT-B-16, and Swin-S, and consider four adversarial attack methods: FGSM, PGD, MI-DI-FGSM, and MI-DI-PGD. First, we conduct self-model attacks, where the surrogate and target models are identical, allowing us to compare the effectiveness of the four attack methods on each

detector. Based on these results, we select the most effective method and use it for subsequent experiments involving cross-model transferability and ensemble attacks.

All detectors were trained using the common training protocol described in Section II. Unless otherwise specified, the adversarial evaluations in Tables 1 and 2 were conducted at an SNR of 10 dB with a perturbation budget of $\epsilon = 16/255$. For Fig. 2, the perturbation budget was varied from $2/255$ to $16/255$ to examine attack behavior across different perturbation strengths. The norm constraint followed the formulation of each attack method, as described in Section III: FGSM and MI-DI-FGSM used an L_∞ constraint, whereas PGD and

MI-DI-PGD used an L_2 constraint.

We evaluate attack performance using two primary metrics. The first is the ASR, which measures the proportion of samples that were correctly classified before the attack but

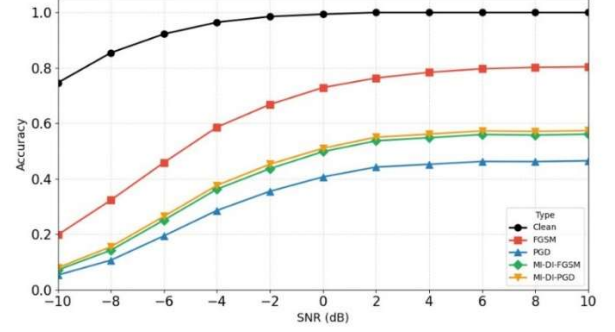


Fig. 3. Detection accuracy versus SNR for adversarial examples across all four detector architectures misclassified afterward, indicating the direct effectiveness of each adversarial method. The second is cross-model transferability, which quantifies how well perturbations generated on one surrogate model can degrade the performance of other target models, thereby reflecting the generalization ability of the attack across different detector architectures.

Fig. 2 illustrates the average ASR as a function of the perturbation magnitude for four different attack methods (FGSM, PGD, MI-DI-FGSM, and MI-DI-PGD) under the self-model attack setting. That is, each adversarial example was generated and evaluated on the same model, and the results were averaged across all four models (ResNet50, MobileNetV2, ViT-B-16, and Swin-S) to provide a model-agnostic comparison. The plot reveals that PGD consistently achieves the highest ASR across all ϵ levels, confirming its effectiveness as

a multi-step iterative attack. MI-DI-FGSM and MI-DI-PGD follow closely, benefiting from momentum and input diversity strategies, but remain slightly less effective than standard PGD in this setting. FGSM, being a single-step method, shows the lowest ASR results.

Because the four methods employ different norm constraints according to their underlying formulations, Fig. 2 should be interpreted

primarily as an empirical comparison of the attack methods under the perturbation settings adopted in this study, rather than as a strictly norm-equivalent comparison of optimization strength. Nevertheless, the results show that iterative optimization provides substantially stronger self-model attack effectiveness than the single-step FGSM baseline

Table 1. ASR(%) under single-model PGD attacks across all combinations of attack and detection models.
($T=20$ steps, $\epsilon=16/255$, l_2 constraint)

		Detect Model			
		ResNet50	Mobile NetV2	ViT (B-16)	Swin-S
Attack Model	ResNet50	96.07	34.87	1.37	3.63
	Mobile NetV2	3.25	99.98	0.88	2.05
	ViT (B-16)	7.68	27.66	85.43	10.08
	Swin-S	5.59	31.04	2.66	98.26

under the evaluated conditions.

While MI-DI-FGSM achieves higher ASR than FGSM due to momentum and input diversity, MI-DI-PGD does not necessarily outperform vanilla PGD. This behavior is consistent with the different objectives of the additional MI and DI mechanisms: rather than solely maximizing attack effectiveness against the source model, they are intended to reduce source-model-specific optimization and improve the generalization of adversarial perturbations. Consequently, the strongest self-model ASR does not necessarily imply the strongest transferability.

Fig. 3 illustrates the classification accuracy of LPI radar detectors across varying SNR levels under different attack methods. The results are averaged across four detector architectures, and each curve represents a specific attack setting, including the clean (non-adversarial) baseline for comparison. The black curve (Clean) represents the accuracy of detectors without any adversarial perturbation. As expected, accuracy improves steadily with increasing SNR, exceeding 90% in high-SNR regimes, indicating that all models perform reliably under clean conditions.

In contrast, the presence of adversarial perturbations causes significant degradation in detection performance across all SNR levels. PGD exhibits the most damaging effect, followed closely by MI-DI-FGSM and MI-DI-PGD, while FGSM yields the least degradation among the four attack methods.

The degradation persists even as the clean classification accuracy increases at higher SNRs, indicating that the observed adversarial vulnerability is not limited to conditions in which the underlying radar signal is already difficult to classify. These results provide the basis for selecting PGD for the subsequent transferability analysis.

Following the comparison of adversarial methods in Figs 2. and 3., we fix PGD as the attack strategy for subsequent evaluations due to its superior performance across models and noise levels.

Table 1 reports the ASR of single-model PGD attacks at distinct cases to be separated. The diagonal entries correspond to self-model attacks, in which the adversarial examples are generated and evaluated on the exact same trained model. These entries therefore represent white-box attack effectiveness and are not interpreted as black-box transfer results. In contrast, the off-diagonal entries correspond to transfer attacks, in which adversarial examples generated using one model are evaluated on a different detector without using the target detector's gradients during adversarial generation.

The diagonal self-model attacks achieve ASRs of 96.07%, 99.98%, 85.43%, and 98.26% for ResNet50, MobileNetV2, ViT-B-16, and Swin-S, respectively. These high values confirm that PGD is highly effective when gradients of the exact target model are available. However, transferability decreases substantially when the generated perturbations are applied to a different detector. Several off-diagonal ASRs fall below 5%, demonstrating that perturbations optimized for a specific source model frequently fail to generalize to another

architecture.

A closer examination of the off-diagonal entries reveals that adversarial examples generated by MobileNetV2 have the lowest transferability, achieving an average ASR of only 2.06% across other models. In contrast, those generated by ViT-B-16 transfer most effectively, with an average ASR of 15.14%. From a robustness perspective, MobileNetV2 is also the most vulnerable as a target detector, with an average ASR of 31.19% under attacks from other models. Conversely, ViT-B-16 demonstrates the highest

Table 2. ASR(%) of ensemble-model PGD attacks (2-, 3-, and 4-model ensembles; $T=20$ steps, $\epsilon = 16/255$, l_2 constraint, Logit-mean)

Results for each detection architecture are averaged over five independently configured target models.

Attack Models (Ensemble)	Detection Model				Avg
	ResNet50	MobileNetV2	ViT (B-16)	Swin-S	
Combination					
ResNet50 MobileNetV2	39.94	99.79	1.68	4.17	36.40
ResNet50 ViT-B-16	55.37	35.75	37.06	9.38	34.39
ResNet50 Swin-S	55.37	35.39	2.90	78.76	42.68
MobileNetV2 ViT-B-16	6.41	99.95	27.98	7.31	35.41
MobileNetV2 Swin-S	5.95	99.97	2.71	56.47	41.28
ViT-B-16 Swin-S	9.77	32.21	48.44	95.09	46.38
ResNet50 MobileNetV2 ViT-B-16	40.29	99.80	26.15	9.08	43.83
ResNet50 MobileNetV2 Swin-S	41.15	99.78	3.11	50.38	48.61
ResNet50 ViT-B-16 Swin-S	53.31	35.82	34.76	77.11	50.25
MobileNetV2 ViT-B-16 Swin-S	9.48	99.94	29.30	54.88	48.40
ResNet50 MobileNetV2 ViT-B-16 Swin-S	40.86	99.79	26.99	49.44	54.27

robustness, with an average ASR of only 1.64%, highlighting its strong resistance to transferred perturbations.

Importantly, the diagonal and off-diagonal entries represent different attack settings. The diagonal entries correspond to self-model

attacks, whereas the off-diagonal entries represent single-surrogate transfer attacks. The mean ASR of the 12 off-diagonal cases is 10.90%, which is used as the single-model transfer baseline in the subsequent analysis. SNR = 10 dB and $\epsilon = 16/255$. In this experiment, the four fixed models serve directly as both attack and detection models, allowing two distinct cases to be separated.

These results underscore the importance of architectural alignment between surrogate and target models in black-box attack settings and motivate the need for more robust, transferable attack strategies such as ensemble-based approaches, which are the main focus of the

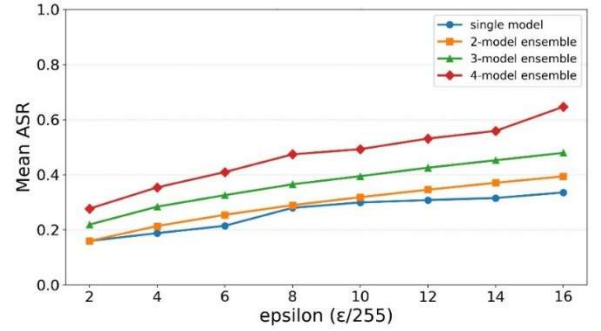


Fig. 4. Comparison of single-model and ensemble attacks under varying ϵ (PGD based) subsequent analysis.

To improve the transferability of adversarial examples across heterogeneous detector architectures, we further apply ensemble-based attack generation. Table 2 presents the ASR results of PGD-based ensemble attacks across different detector models. The table lists the ASR achieved for each specific combination of surrogate models, including all six pairwise combinations for 2-model ensembles, four triple combinations for 3-model ensembles, and the full 4-model ensemble. These results reveal how individual model combinations influence attack transferability. As described in Section II, the exact target detector is not included in adversarial generation. For further analysis, we distinguish between same-family-included cases, where the ensemble contains a surrogate from the same architecture family as the target, and same-family-excluded cases, where the target architecture family is absent from the ensemble. Compared to the off-diagonal

single-model transfer results in Table 1, ensemble-based attacks generally improve adversarial transferability, although the degree of improvement depends on the architectural relationship between the surrogate ensemble and the target detector.

Specifically, the mean ASR is 61.36% for same-family-included cases and 13.20% for same-family-excluded cases. The latter remains slightly higher than the 10.90% mean ASR of the off-diagonal single-surrogate cases in Table 1, indicating that ensemble generation can improve transferability even when the target architecture family is not represented among the surrogates.

Notably, the four-model ensemble attack yields the highest average ASR of 54.27% across all

	BPSK	Costas	Frank	LFM	P1	P2	P3	P4	T1	T2	T3	T4
BPSK	34.4 (52.9)	6.3 (4.9)	2.6 (1.7)	1.1 (0.0)	2.1 (1.5)	3.1 (1.4)	1.3 (0.0)	0.0 (0.0)	26.6 (23.9)	16.9 (10.7)	1.3 (0.0)	3.4 (1.5)
Costas	20.3 (15.7)	40.7 (67.6)	2.7 (3.5)	3.5 (1.9)	4.3 (2.7)	3.4 (1.8)	7.2 (1.4)	4.1 (2.0)	6.8 (1.3)	2.9 (1.1)	1.7 (0.0)	2.2 (0.0)
Frank	12.3 (3.4)	4.0 (6.7)	19.9 (34.9)	2.3 (0.0)	5.4 (1.3)	0.0 (0.0)	50.0 (50.1)	2.8 (1.1)	1.1 (0.0)	0.0 (0.0)	1.0 (0.0)	0.0 (0.0)
LFM	7.4 (3.1)	0.0 (0.0)	1.7 (2.0)	35.5 (58.6)	13.7 (7.6)	0.0 (0.0)	1.1 (0.0)	31.2 (24.9)	2.6 (1.1)	5.3 (2.1)	0.0 (0.0)	0.0 (0.0)
P1	8.1 (2.2)	2.3 (1.3)	12.1 (5.0)	10.6 (8.1)	25.0 (49.0)	0.0 (0.0)	10.4 (5.5)	26.3 (23.8)	0.0 (0.0)	2.5 (3.3)	0.0 (0.0)	1.5 (0.0)
P2	14.5 (8.9)	9.4 (7.7)	2.3 (3.6)	1.0 (1.3)	4.1 (1.5)	47.8 (69.9)	2.0 (0.0)	0.0 (1.5)	4.5 (0.0)	1.1 (0.0)	2.9 (1.5)	9.4 (3.0)
P3	5.3 (2.0)	0.0 (0.0)	73.9 (70.3)	0.0 (0.0)	2.6 (2.0)	0.0 (0.0)	12.8 (24.6)	2.5 (0.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)
P4	6.2 (1.3)	1.6 (1.4)	6.3 (5.6)	14.8 (15.7)	34.2 (30.5)	0.0 (0.0)	8.0 (0.0)	25.0 (43.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	2.8 (0.0)
T1	29.3 (23.8)	0.0 (0.0)	2.7 (1.9)	1.7 (1.1)	2.0 (0.0)	1.9 (1.8)	0.0 (0.0)	1.1 (0.0)	40.0 (63.0)	9.4 (2.9)	2.9 (2.1)	7.9 (2.2)
T2	28.9 (21.3)	0.0 (0.0)	0.0 (0.0)	2.1 (0.0)	1.2 (0.0)	2.3 (1.9)	1.9 (0.0)	10.6 (1.4)	40.2 (5.5)	40.2 (61.3)	3.6 (1.9)	7.5 (4.6)
T3	21.7 (10.8)	1.1 (1.4)	7.6 (11.3)	1.5 (0.0)	1.6 (1.1)	2.8 (1.5)	4.1 (2.4)	1.7 (1.2)	7.0 (3.8)	2.5 (2.1)	41.8 (61.0)	6.6 (2.7)
T4	9.5 (2.9)	0.0 (0.0)	1.2 (0.0)	2.4 (0.0)	10.1 (3.7)	8.9 (3.5)	0.0 (0.0)	11.2 (17.3)	5.2 (2.5)	6.6 (4.4)	4.4 (3.4)	39.1 (60.3)

Fig. 5. Per-class accuracy under ensemble PGD attacks (averaged across ϵ); single-model transfer results shown in parentheses.

target detectors. However, the average ASR of 29.41% in Table 1 includes both self-model and off-diagonal transfer cases and therefore does not provide a direct baseline for black-box transferability. These results indicate that ensemble-based adversarial generation can improve cross-model transferability, while the improvement is substantially greater when the surrogate ensemble includes a model from the same architecture family as the target detector.

Following Table 2, which quantitatively demonstrated differences in transferability according to surrogate ensemble composition,

Fig. 4 provides a visual comparison of attack performance between single-model and ensemble-model PGD attacks across varying perturbation magnitudes ϵ .

As shown, ensemble attacks consistently yield higher ASRs than their single-model counterparts, with the four-model ensemble achieving the most significant improvement. However, as discussed in the analysis of Table 2, the four-model ensemble includes a surrogate from each of the four architecture families considered in this study. Therefore, its high ASR reflects not only the use of multiple surrogate models but also the architectural coverage of the target detector families. These results further indicate that combining diverse surrogate architectures can enhance adversarial transferability, while the degree of improvement depends on the relationship between the surrogate ensemble and the target architecture.

To further analyze how adversarial perturbations affect classification behavior at a finer granularity, we present a confusion matrix in Fig. 5. which presents a detailed confusion matrix illustrating the classification results for twelve LPI radar modulation schemes under adversarial conditions. Each entry shows the classification accuracy from a true class (column) to a predicted class (row). The outer values indicate accuracy under the four-model ensemble PGD attack, while the values inside parentheses correspond to results under single-model PGD attack (self-model case). Diagonal entries represent correct predictions, and off-diagonal entries represent misclassifications. Because these two cases represent different attack settings, the comparison is used primarily to examine changes in class-level error patterns rather than as a direct measure of relative attack strength.

Overall, we observe substantial differences in class-level classification behavior between the single-model and ensemble attack settings. For example, the diagonal accuracy for the Costas

waveform drops from 67.6% single-model to 40.7% ensemble, showing increased classification degradation under the four-model ensemble setting. At the same time, misclassifications become more structured and frequent. For instance, BPSK signals show a marked increase in confusion with the T1 and T2 classes under ensemble attacks. The misclassification rates rose from 23.9% to 26.6% for T1, and from 10.7% to 16.9% for T2. Additionally, new misclassification patterns emerged toward LFM, P3, P4, and T3, which were not observed under single-model attacks. These results indicate that ensemble attacks not only affect class-wise accuracy but also alter class-level confusion patterns driven by shared characteristics in the time and frequency domains.

In summary, Fig. 5 highlights how adversarial examples generated under the ensemble setting produce distinct and pronounced class-level classification errors compared with the single-model setting. Such modulation-aware vulnerability patterns are valuable for designing more robust training strategies and targeted defenses.

Overall, the empirical results demonstrate that adversarial perturbations not only degrade classification accuracy but also introduce structured misclassification patterns, especially under ensemble attacks. The transferability results further show that attack effectiveness is strongly influenced by the architectural relationship between the surrogate ensemble and the target detector. This underscores the need for architecture-aware defense mechanisms and highlights the vulnerabilities inherent in current deep-learning-based LPI radar detectors.

V. Conclusion

This paper systematically investigated the adversarial robustness and cross-model transferability of deep learning-based LPI radar

detectors. We examined four representative architectures—ResNet50, MobileNetV2, ViT-B-16, and Swin-S—spanning both convolutional and Transformer families. Our study compared the effectiveness of four gradient-based attack methods and revealed that PGD achieved the highest overall attack effectiveness under the evaluated conditions and was therefore adopted for the subsequent transferability analysis.

To address the limited transferability of single-model adversarial examples, we introduced ensemble-based attacks that combine information from multiple surrogate models. The results showed that ensemble-based transferability strongly depends on the architectural relationship between the surrogate ensemble and the target detector. The mean ASR increased from 10.90% for off-diagonal single-surrogate attacks to 13.20% for same-family-excluded ensemble attacks, while same-family-included ensemble attacks achieved a substantially higher mean ASR of 61.36%. The four-model ensemble achieved an average ASR of 54.27% across all target detectors; however, this ensemble included surrogates from all four architecture families considered in this study. Furthermore, modulation-wise confusion analysis showed that ensemble attacks not only affect classification accuracy but also produce distinct inter-class confusion patterns, especially for structurally similar modulation types.

These findings highlight the importance of considering architectural diversity and architectural coverage in both attack and defense strategies. For practical LPI radar systems deployed in adversarial environments, defense mechanisms must be evaluated not only against white-box attacks but also under more realistic black-box scenarios involving transferable ensemble attacks. It should also be noted that the adversarial perturbations considered in this study are generated in the TFI domain, and their direct physical correspondence to realizable IQ-domain perturbations remains a limitation of the

present work. Future work may explore adversarial training, model ensembling, or modulation-specific regularization to improve the robustness of LPI radar detectors under real-world conditions, as well as investigate the physical realizability of TFI-domain adversarial perturbations in the original signal domain.

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Moohyun Lee



received the B.E. degree in electrical engineering from Korea Military Academy, South Korea, in 2017, the M.S. degree in electrical engineering from Korea Advanced Institute of Science and Technology, Korea, in 2025. He is currently a Lecturer at the Korea Military Academy, Seoul, South Korea. His research interests include artificial intelligence and computer vision.

Sungjae Park



received the B.E. degree in electrical engineering from Korea Military Academy, South Korea, in 2017, the M.S. degree in electrical and computer engineering from the University of Florida, U.S., in 2024. He is currently an Assistant Professor at the Korea Military

Academy, Seoul, South Korea. His research interests include computer security, virtualization and artificial intelligence.

Sunghwan Cho



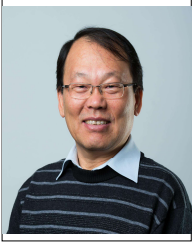
received the D.Phil. degree in Engineering Science from the University of Oxford, U.K., in 2021. He is currently an Associate Professor at the Korea Military Academy, Seoul, South Korea. Since 2007, he has served as an officer in the Republic of Korea Army and currently holds the rank of Lieutenant Colonel. His research interests include visible light communication, physical-layer security, and electrophoretic molecular communications.

Yee Wei Law



received the B.Eng., M.Eng. and Ph.D. degrees from University of Southampton, Nanyang Technological University, and University of Twente respectively. Before joining UniSA, he was a Research Fellow at the Department of Electrical and Electronic Engineering, The University of Melbourne. He is currently a Senior Lecturer at UniSA, focusing on interdisciplinary research related to cybersecurity, machine learning and space.

Sang-Heon Lee



received a Bachelor of EngSci in Aeronautical Eng. from Inha University, Korea, in 1988, a Master of EngSci in Mechatronics from the University of New South Wales in 1995, and a PhD in Systems Engineering from the

Australian National University in 1999. He is currently a Professor in the School of Electrical and Mechanical Engineering at the Adelaide University. His research interests include machine learning for medical and agricultural applications, hyperspectral image processing, and supply chain management. He has published over 150 international peer-reviewed journal and conference papers.

Yongchul Kim



received the B.E. degree in electrical engineering from Korea Military Academy, South Korea, in 1998, the M.S. degree in electrical engineering from the University of Surrey, U.K., in 2001, and the Ph.D. degree in

electrical and computer engineering from the Department of Electrical and Computer Engineering, North Carolina State University, in 2012. He has been a Professor with the Department of Electrical Engineering, Korea Military Academy. His research interests include wireless and satellite communication.